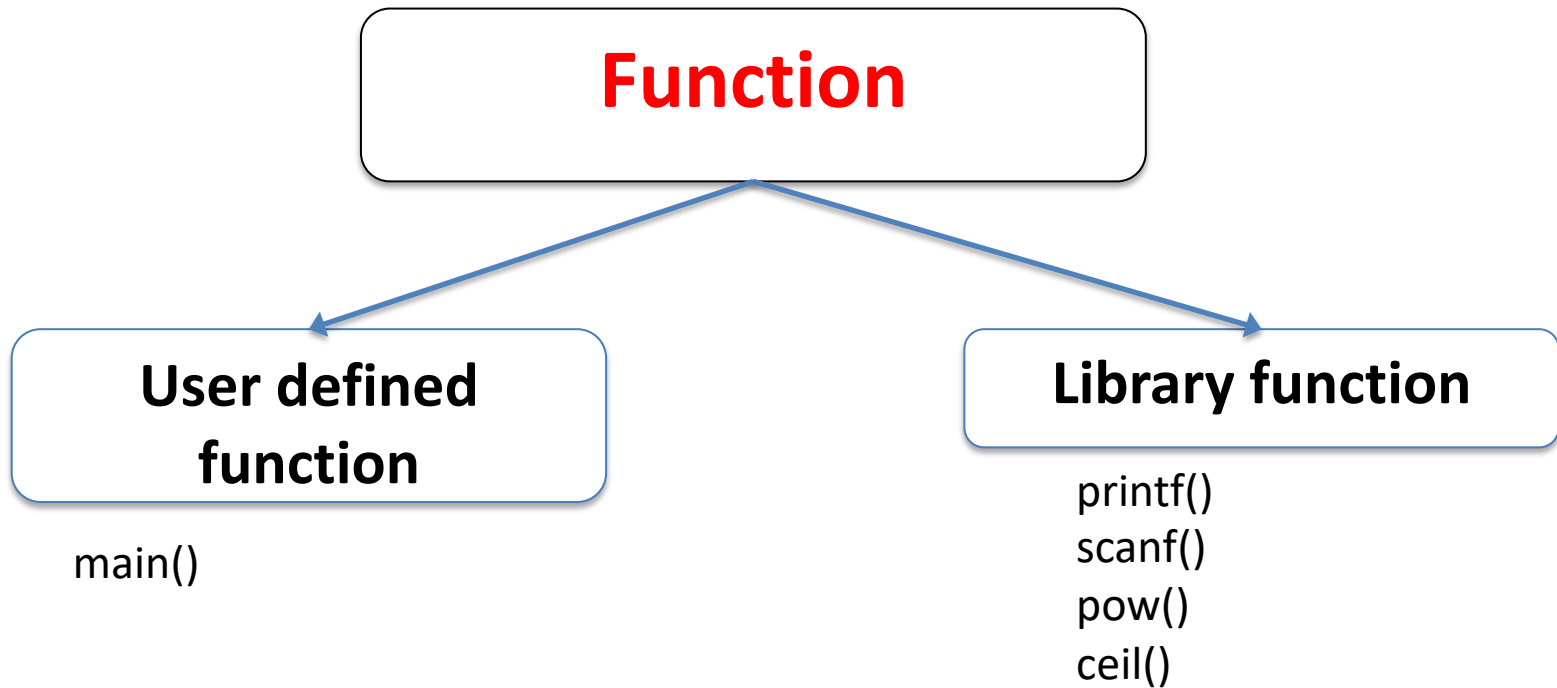


Function

What is function?

- ❖ A large program in C is divided into many subprogram. Each subprogram is called a function.



Example 1

```
#include <stdio.h>

int CubeArea(int x); /* function prototype */

main(){
    int x, area;

    printf("Enter a value: ");
    scanf("%d", &x);

    area = CubeArea(x); /* calling function */

    printf("The area of cube is %d.", area);
    return 0;
}

int CubeArea(int x){ /* function definition */
    int a;

    a = x * x * x;

    return a;
}
```

➤ The variable to be passed from calling function to called function is called argument.

➤ **void** print(int x);

Output

Enter a value: 4
The area of cube is 64.

Example 2

```
#include <stdio.h>

int input(); /* function prototype */

void main(){
    int x;

    x = input(); /* calling function */

    printf("The inputted is %d.", x);
}

int input(){ /* function definition */
    int a;

    printf("Enter a value: ");
    scanf("%d", &a);

    return a;
}
```

Variable Length Arguments

```
int main()
{
    printf("Average of {2, 3, 4} = %d\n", average(2, 3, 4));
    printf("Average of {3, 5, 10, 15} = %d\n", average(3, 5, 10, 15));
    return 0;
}

int average(int num, ...){
    va_list valist;
    int sum = 0, i;

    va_start(valist, num);

    for (i = 0; i < num; i++)
        sum += va_arg(valist, int);

    va_end(valist);

    return sum / num;
}
```

Header file: stdarg.h

Practice

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

```
#include <stdio.h>
```

```
/* function prototype */
```

```
double power(double x, int n);  
long fact(int n);
```

```
/* main function */
```

```
void main(){  
    double x, term = 1, sum = 0;  
    int n = 1;  
  
    .....  
    while ( term > 0.001){  
        .....  
        term = power(x, n)/fact(n);  
        n++;  
    }  
  
    printf("The result is %.2f.", sum);  
}
```

```
/* power function */
```

```
double power(double x, int n){  
    double p;  
    .....  
    return p;  
}
```

```
/* fact function */
```

```
long fact(int n){  
    long f;  
    .....  
    return f;  
}
```

Write down functions for calculating values of $\sin(x)$, $\cos(x)$ and $\log(1 + x)$.